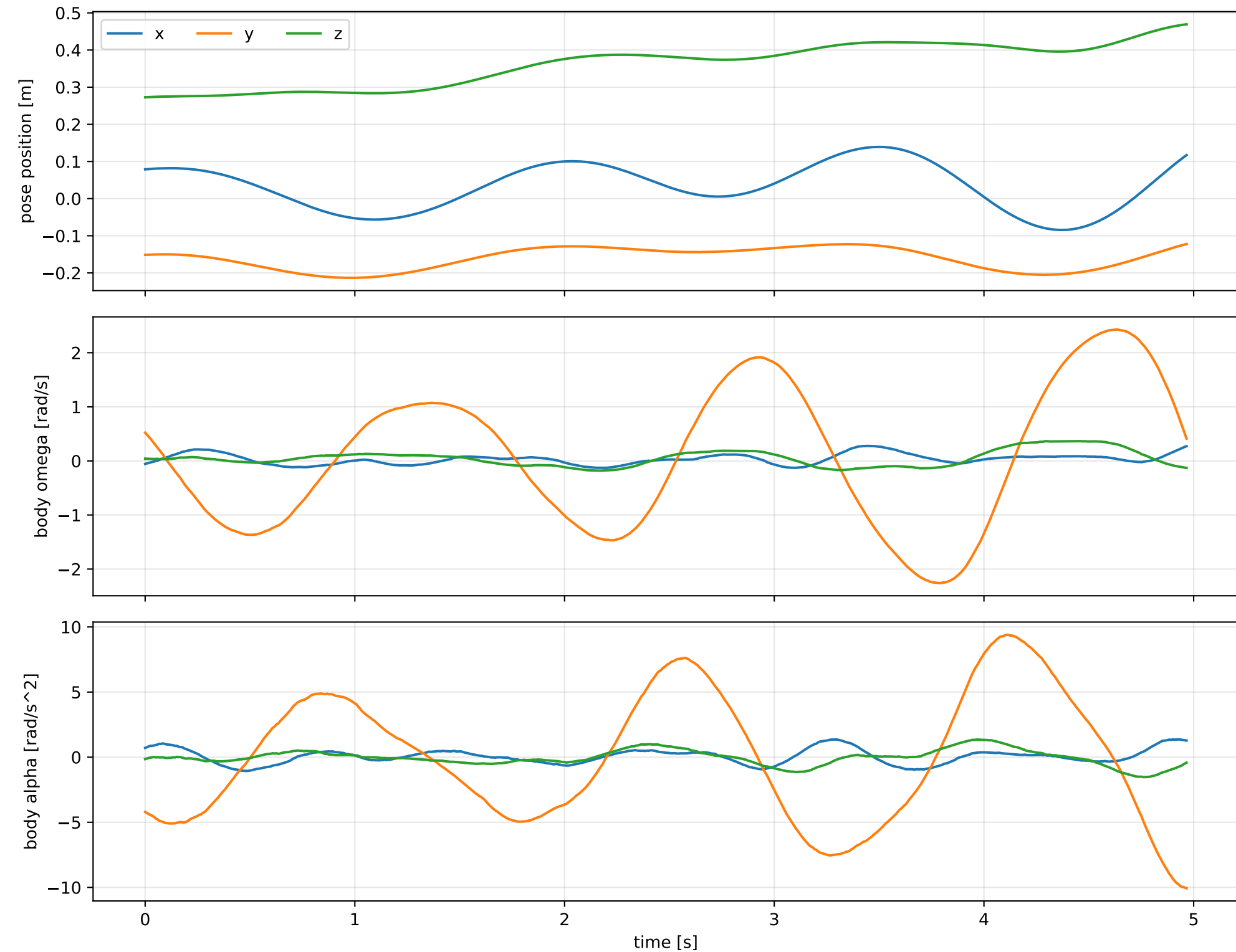
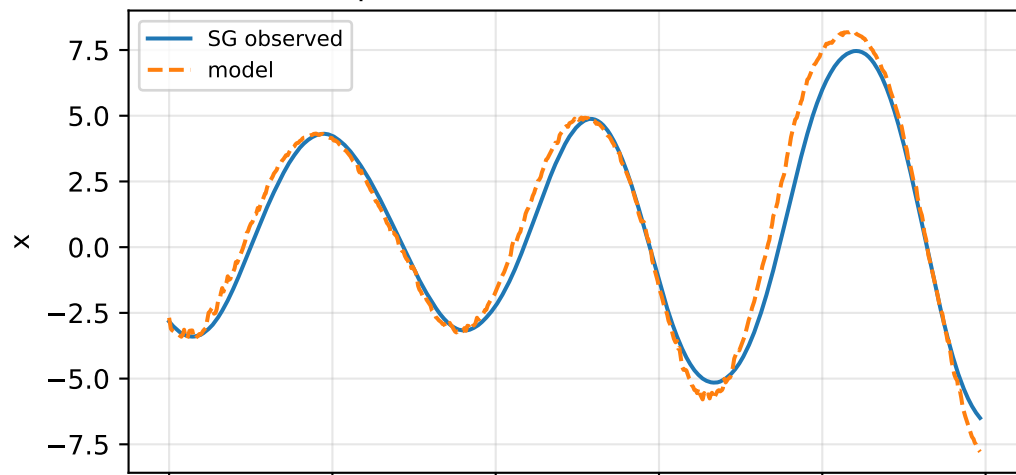


naive_all: SG trajectory and derived motion

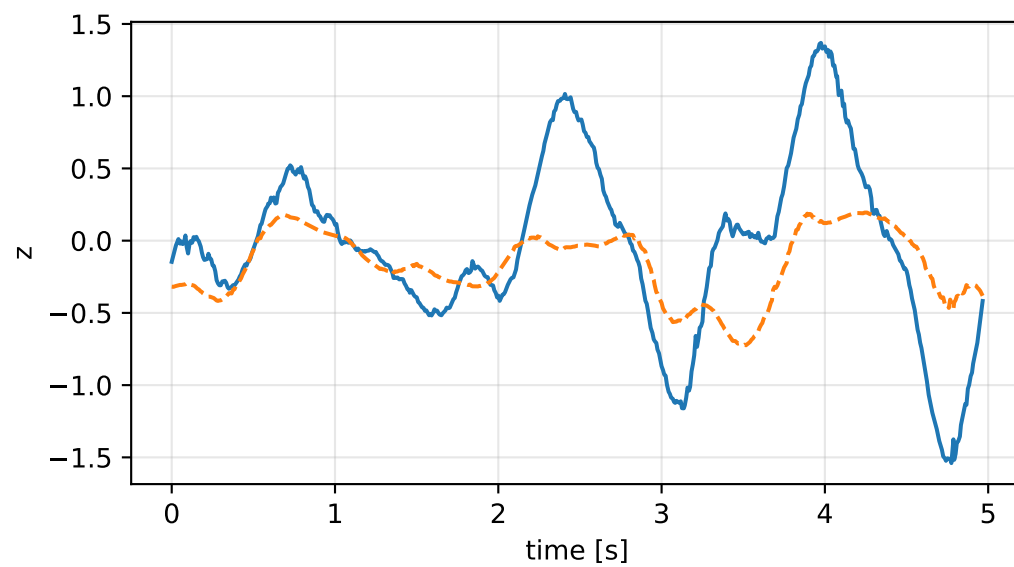
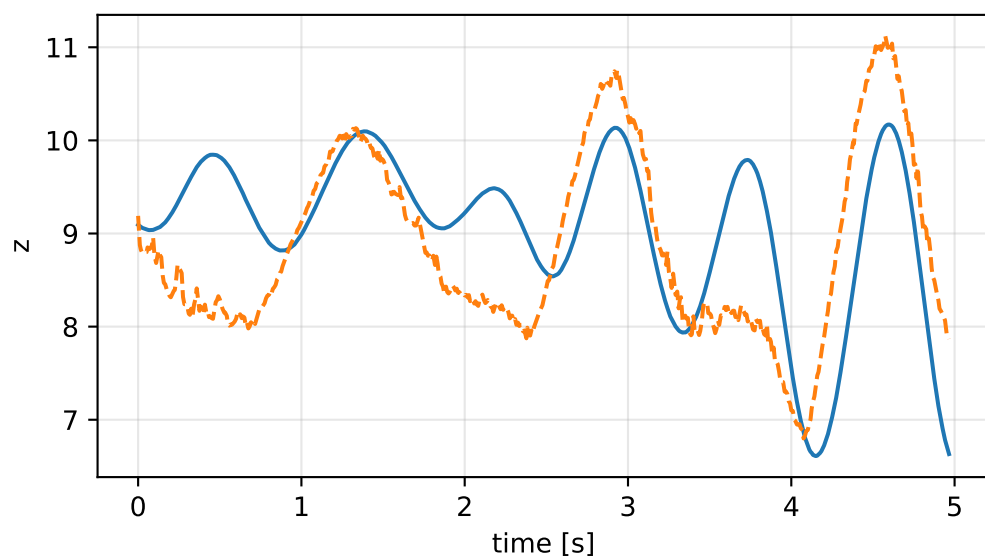
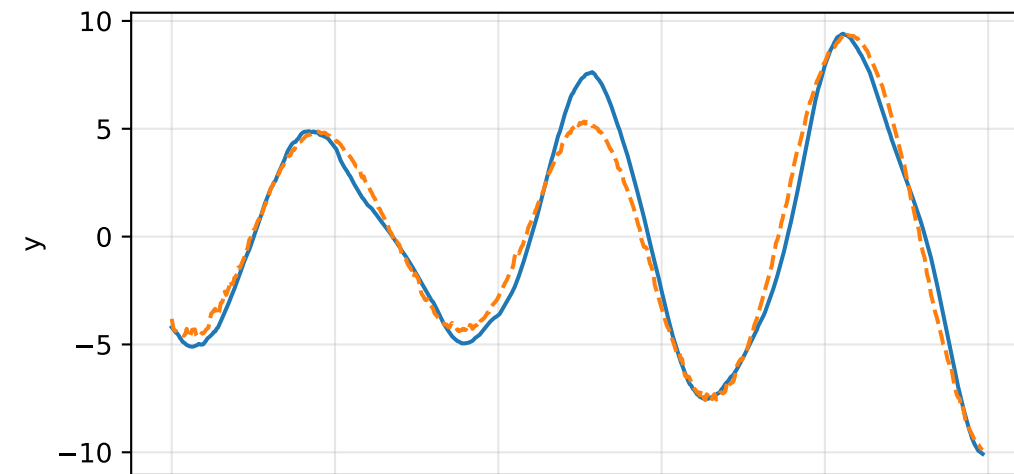
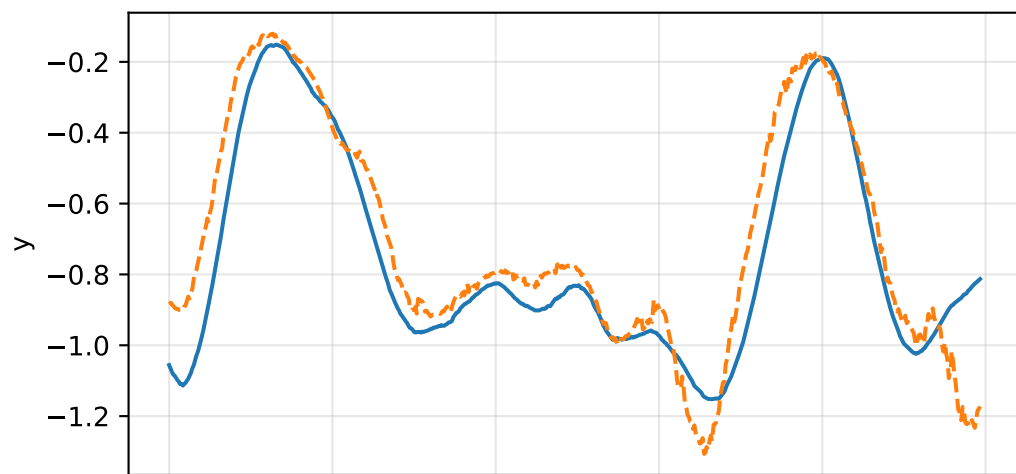
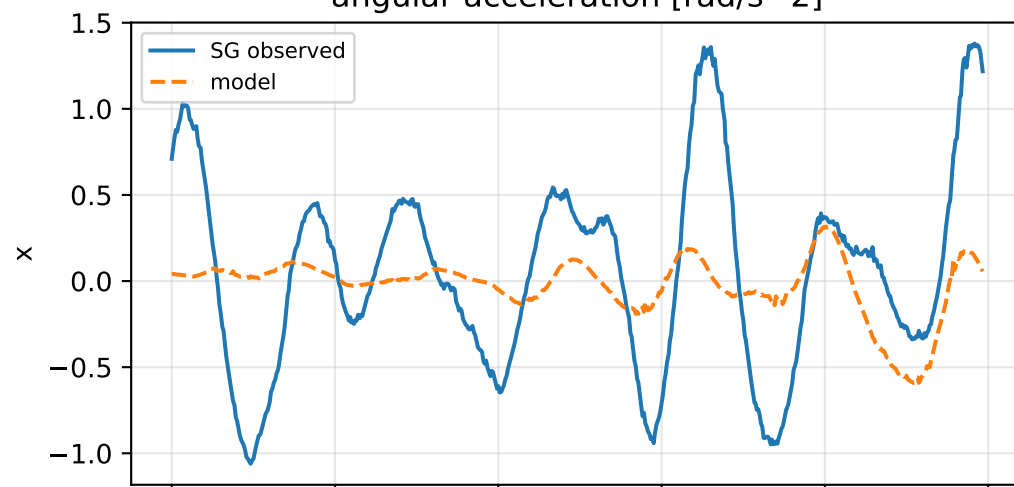


naive_all: acceleration residual objective

specific acceleration [m/s^2]

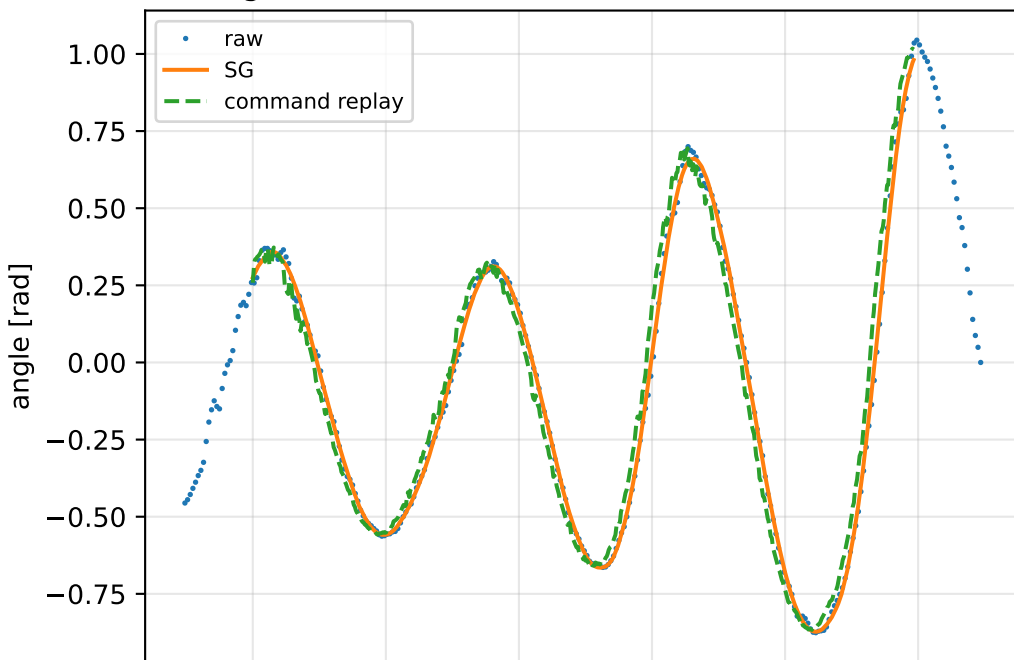


angular acceleration [rad/s^2]

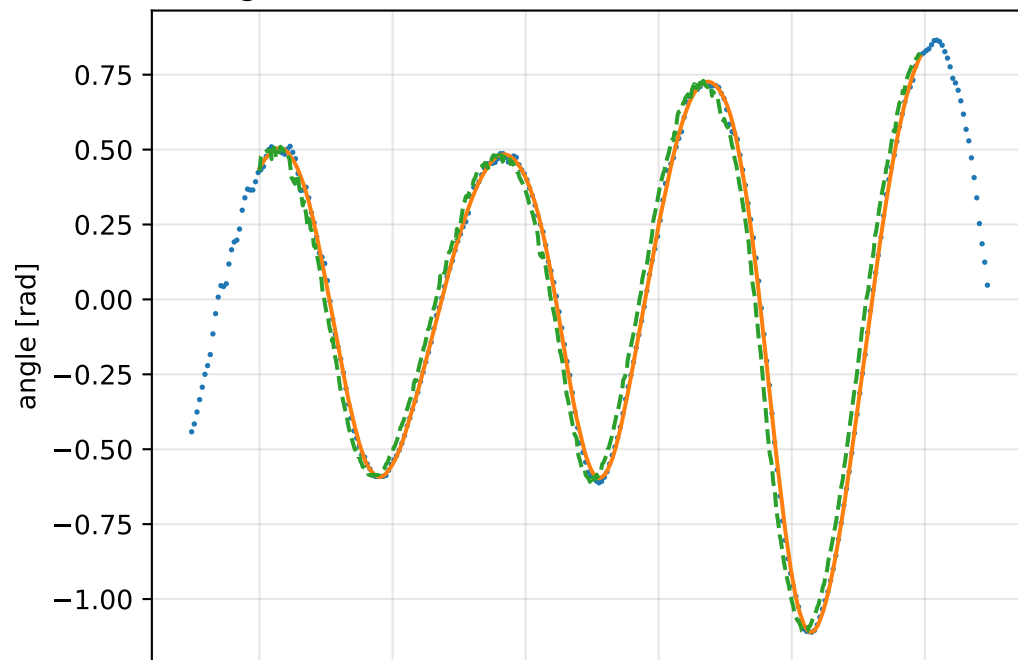


naive_all: gimbal measurement audit (objective=command_replay)

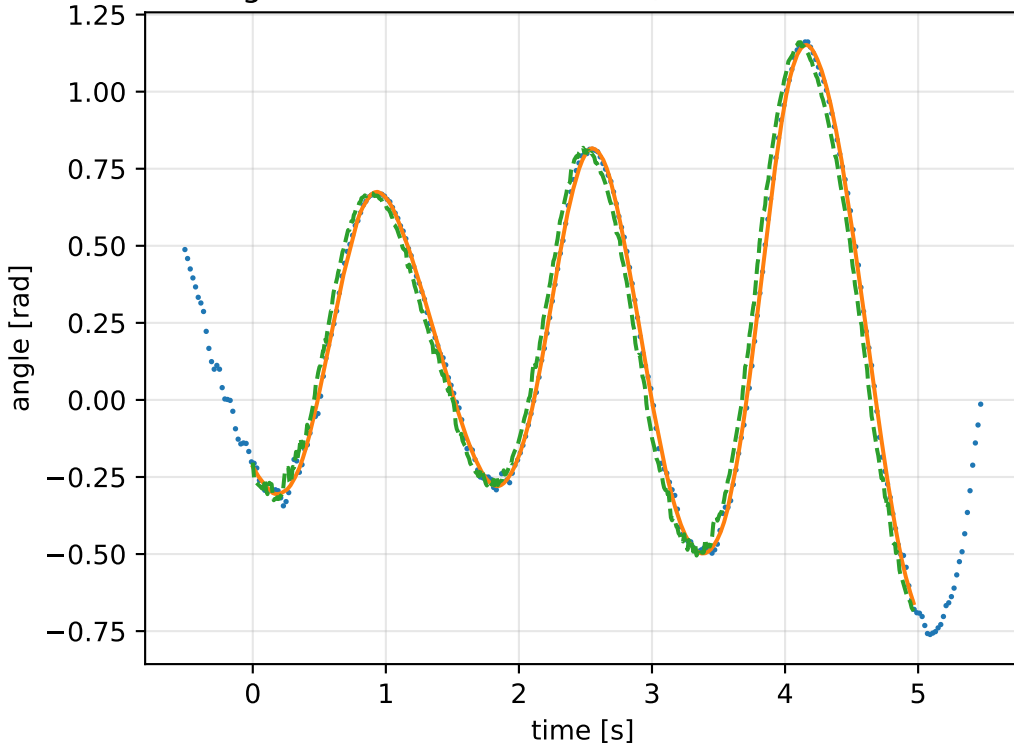
gimbal 1: RMSE=0.0759, max=0.2095 rad



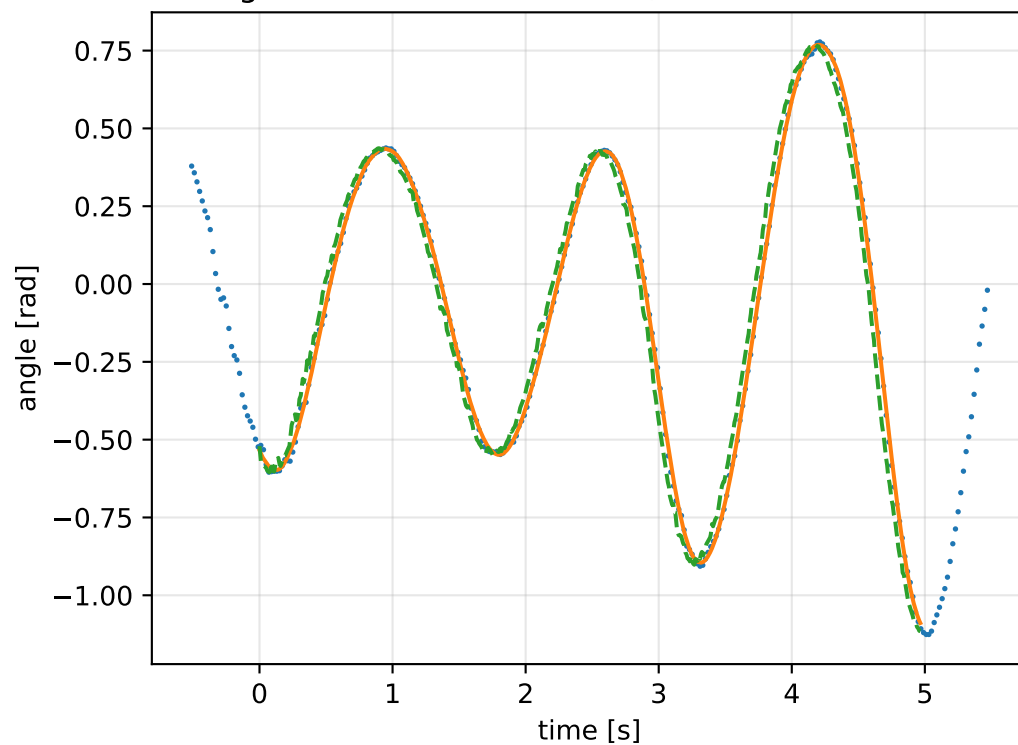
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

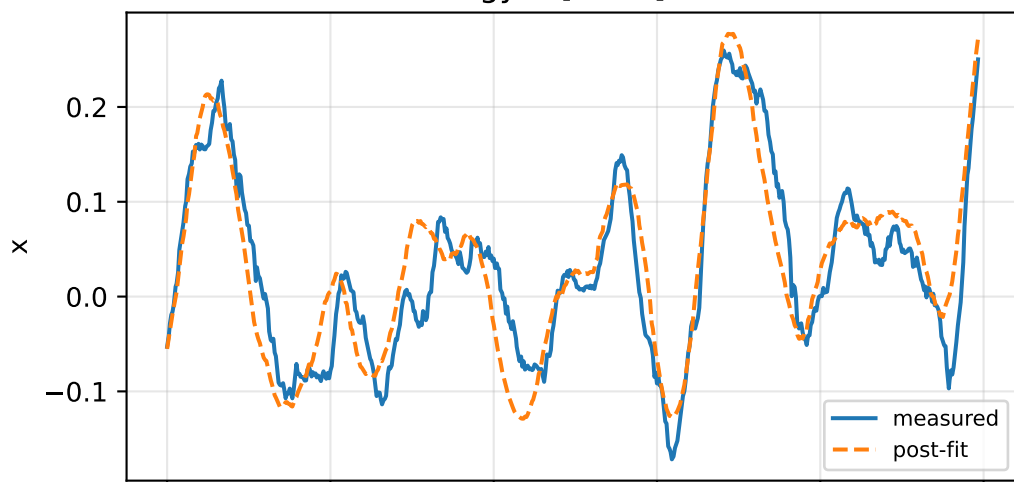


gimbal 4: RMSE=0.07131, max=0.1737 rad

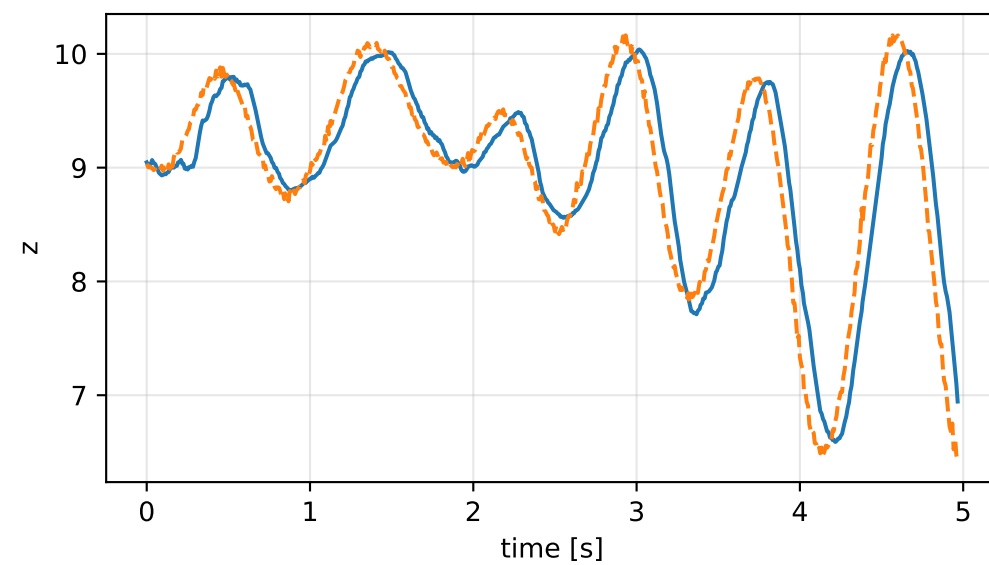
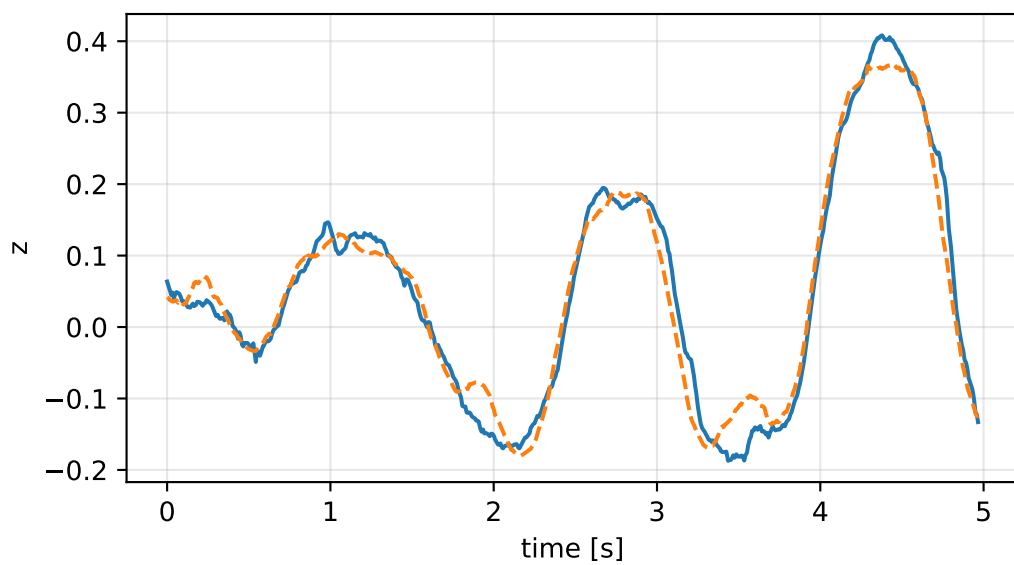
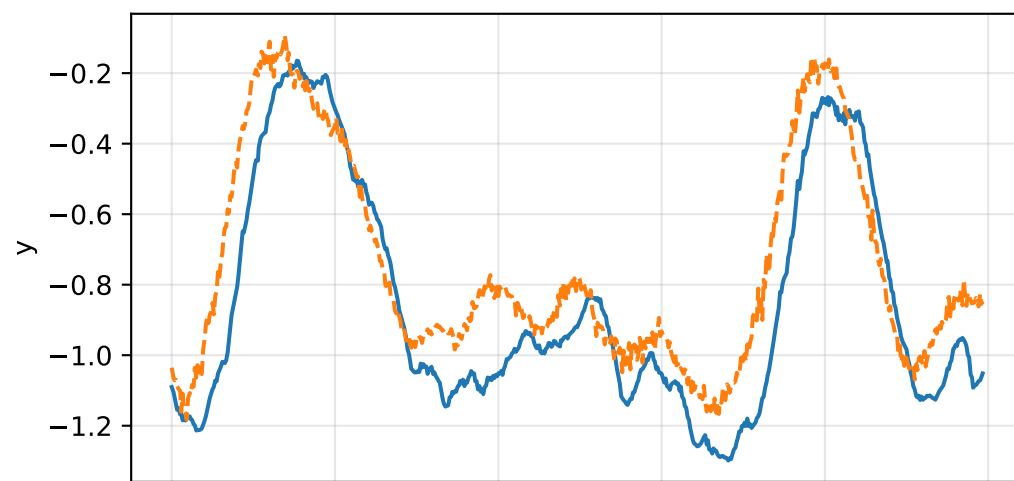
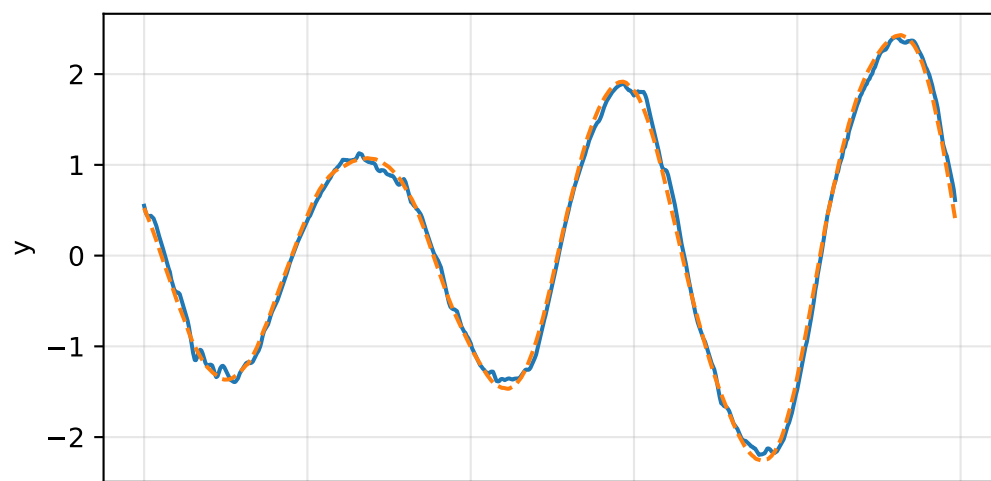
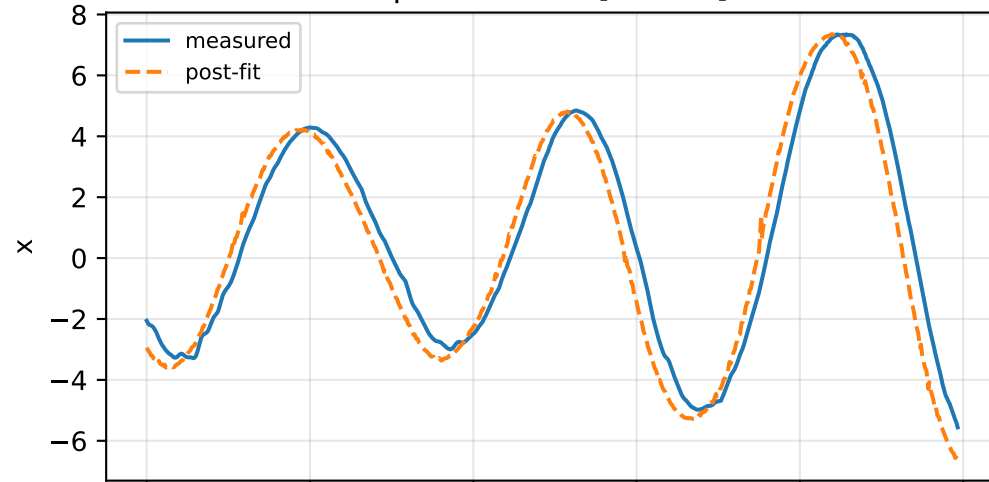


naive_all: IMU comparison (diagnostic only)

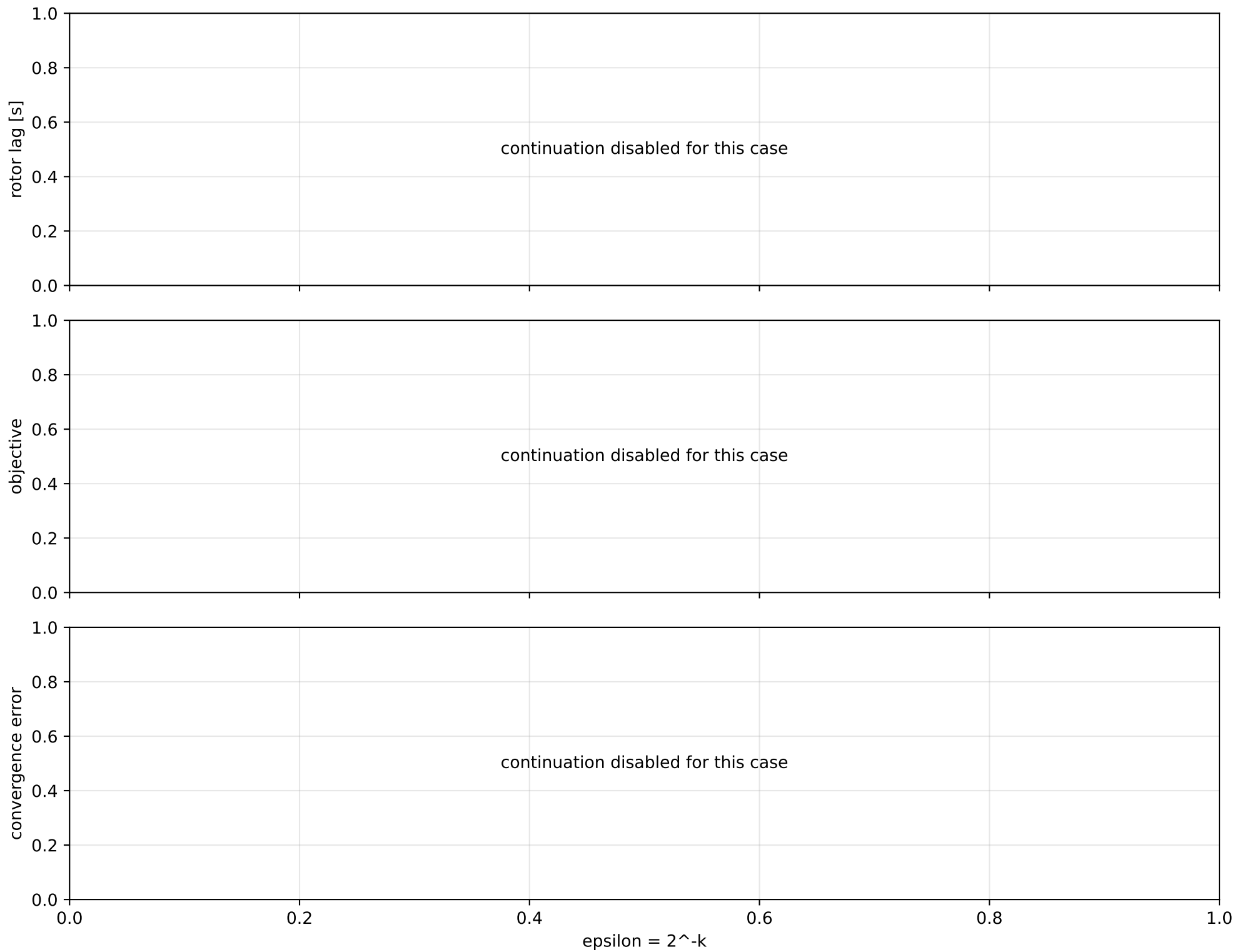
gyro [rad/s]



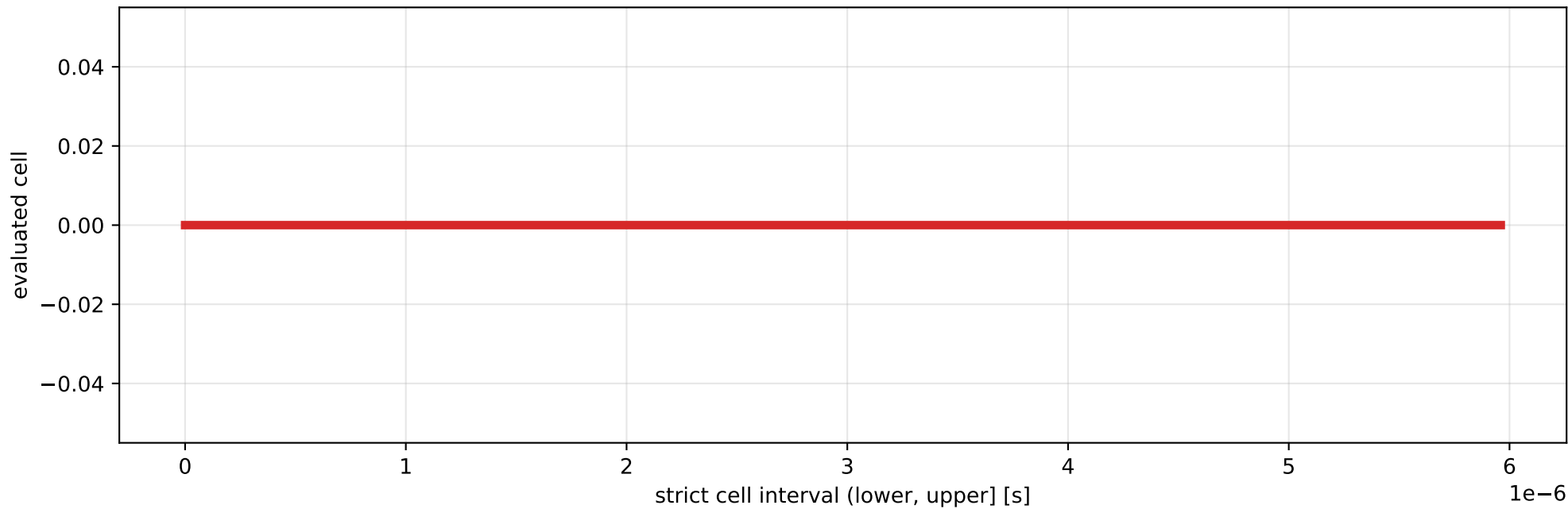
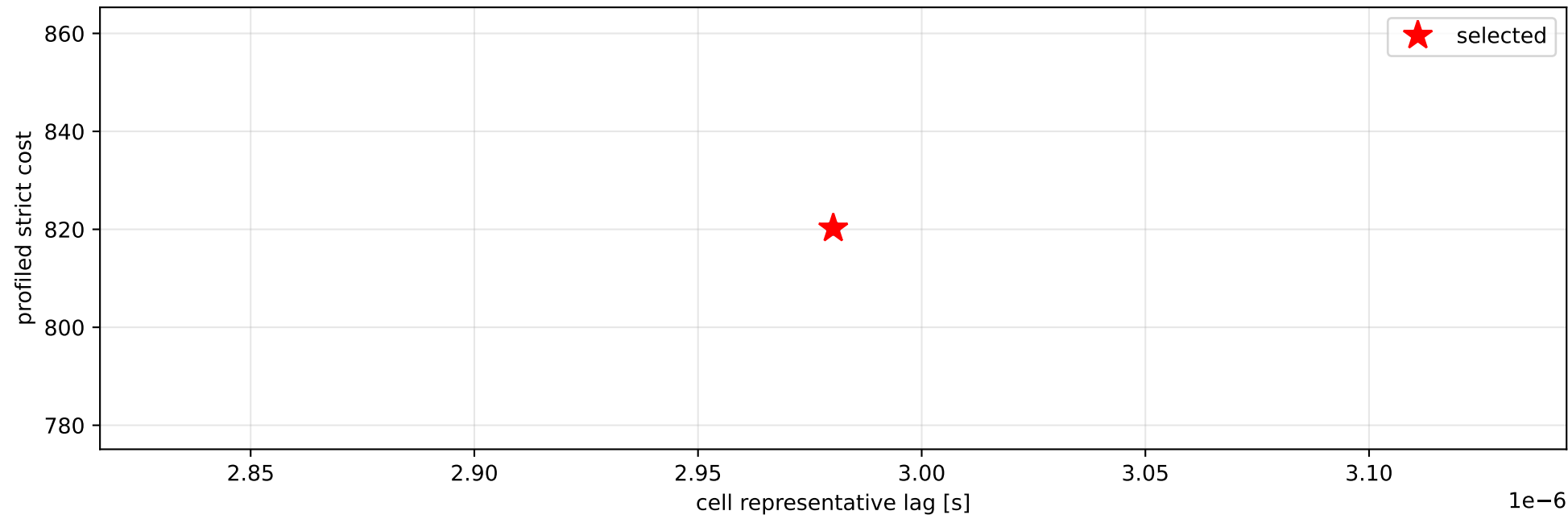
specific force [m/s²]



naive_all: smooth-to-strict continuation



naive_all: exact strict-ZOH lag cells



selected (0, 5.96046448e-06] s; final smooth 0 s; data boundary=False

naive_all: parameter estimate

Case: naive_all

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 28.4075953

inertia [kg m²]:

```
[[33.46438457 -0.22072045  0.16904146]
 [-0.22072045  0.42497993 -1.04354515]
 [ 0.16904146 -1.04354515 33.29289324]]
```

CoG body [m]: [0.22284446 -0.00998886 -0.03661658]

force effectiveness: [7.17844657e-05 2.17816229e+01 1.95982414e+01 7.14644575e-05]

scale-free J/m [m²]:

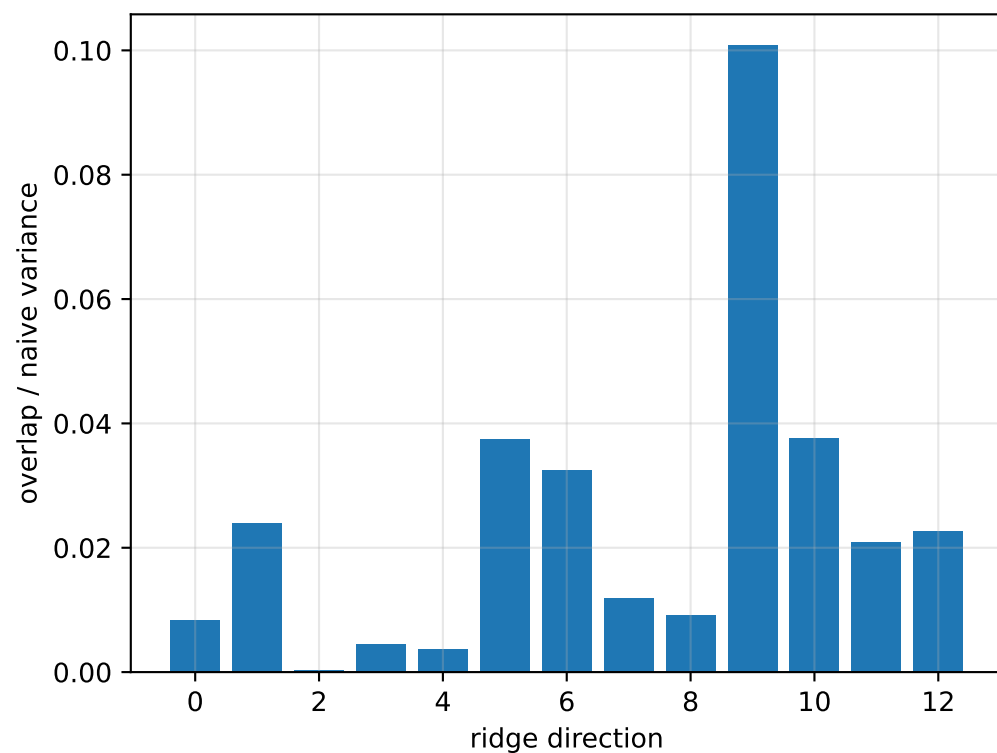
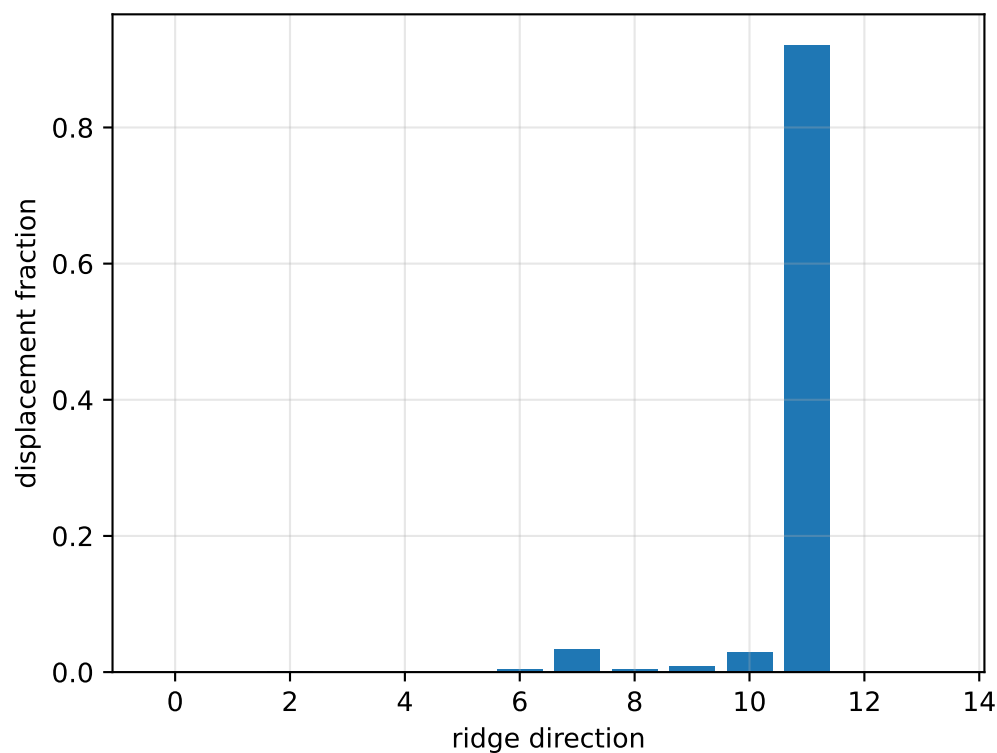
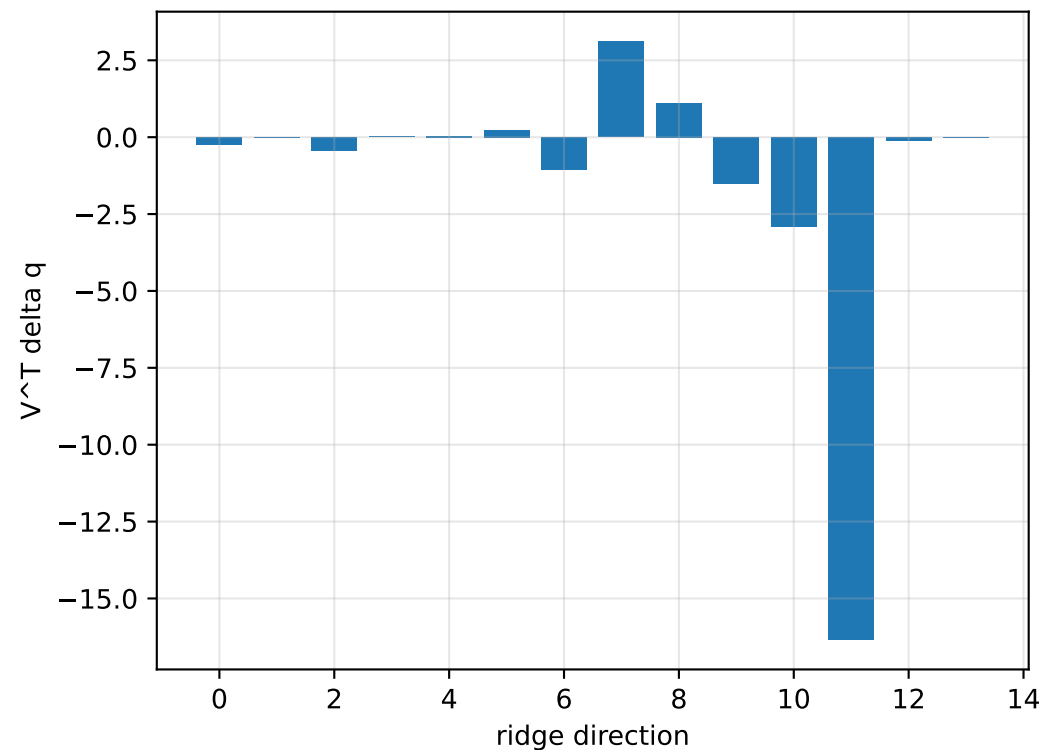
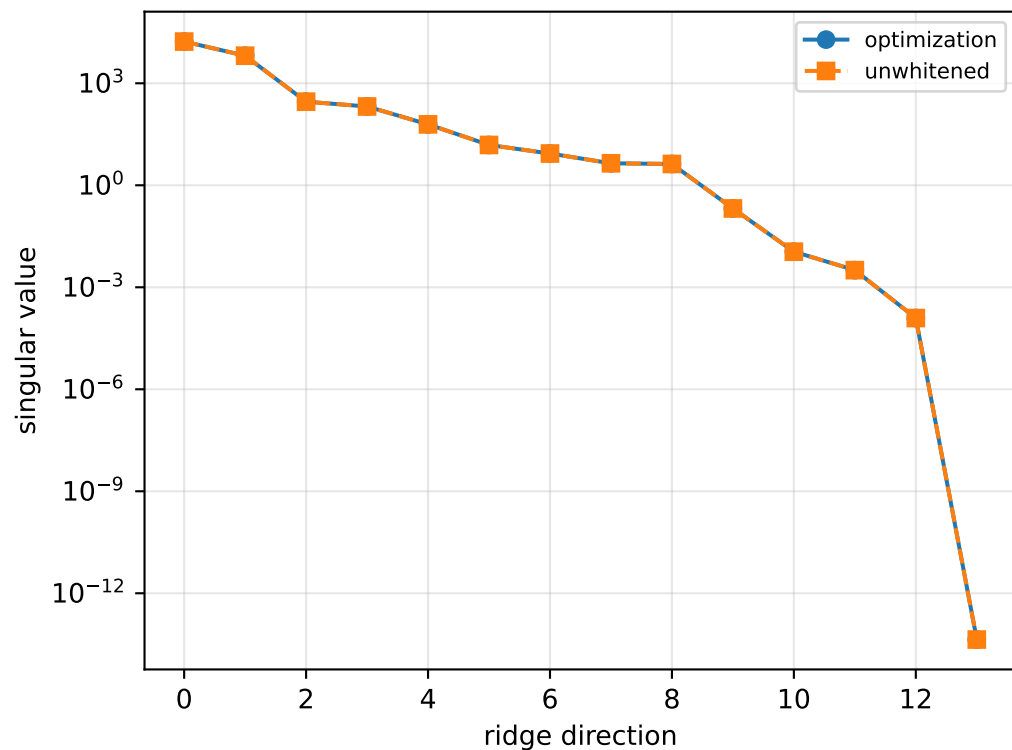
```
[[ 1.17800835 -0.00776977  0.00595057]
 [-0.00776977  0.01496008 -0.03673472]
 [ 0.00595057 -0.03673472  1.17197154]]
```

scale-free f/m: [2.52694623e-06 7.66753494e-01 6.89894419e-01 2.51568134e-06]

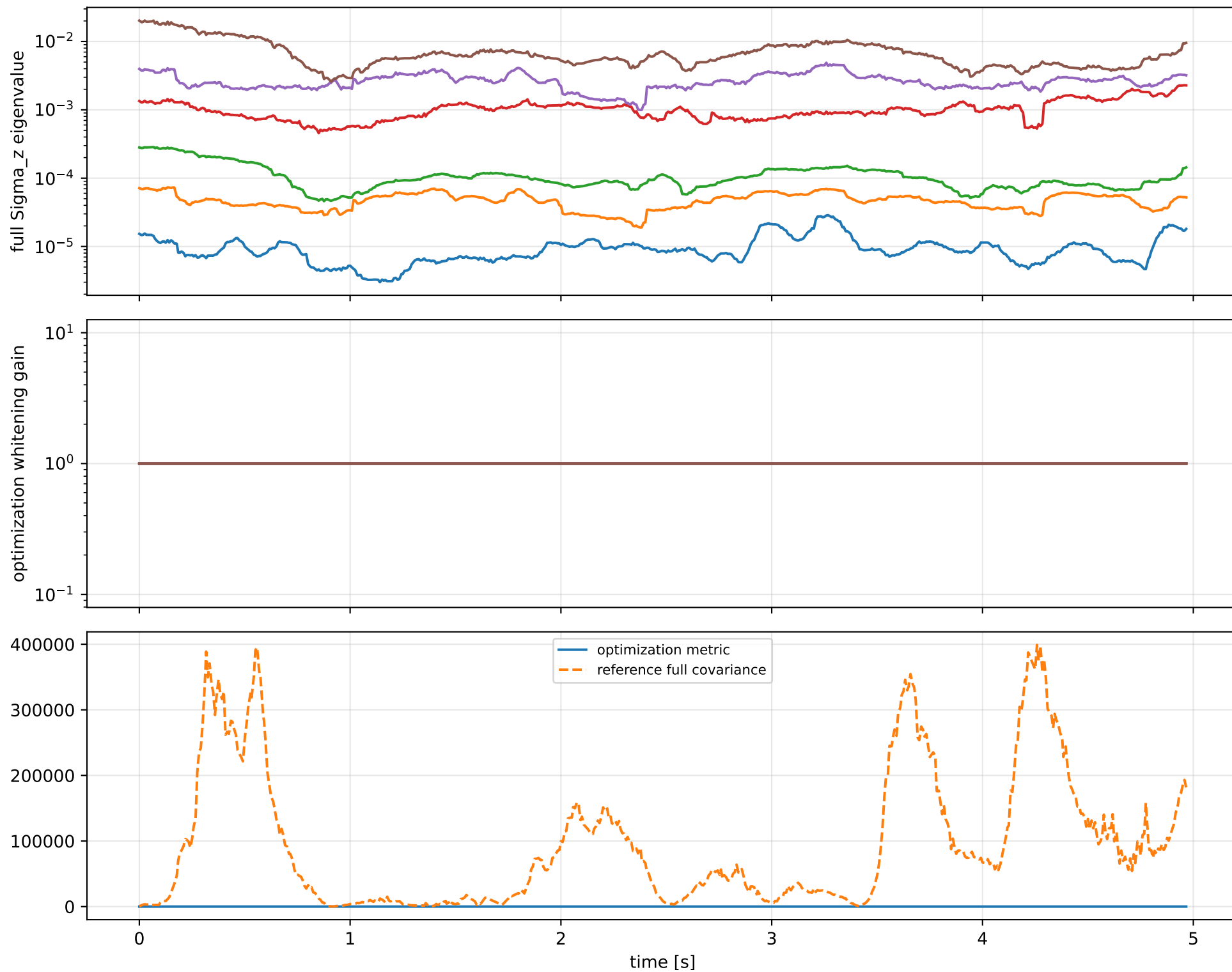
rotor lag cell: (0, 5.96046448e-06] s

representative: 0 s

naive_all: ridge spectrum (rank 13, nullity 1, $||\mathbf{v}||=9.32\text{e-}14$)



naive_all: optimization vs reference covariance



naive_all: wrench / closure (force max 9.99e-14, torque max 1.42e-14)

